

10. (a) (i) Tin and carbon are both more stable in oxidation state +4 whilst lead is more stable as +2. Explain the difference in the stabilities of these oxidation states. [1]

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- (ii) Lead(IV) oxide can act as an oxidising agent when it reacts with concentrated hydrochloric acid. Write an equation for this reaction. [1]

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- (b) Tin(II) ions can be used to reduce iron(III) ions in **acidic** solution.



$[\text{Fe}^{3+}] / \text{mol dm}^{-3}$	$[\text{Sn}^{2+}] / \text{mol dm}^{-3}$	pH	Initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
0.200	0.100	0	1.4×10^{-2}
0.300	0.200	0	2.8×10^{-2}
0.400	0.100	1	1.4×10^{-3}
0.400	0.200	1	2.8×10^{-3}

- (i) Find the rate equation for this reaction. [3]

- (ii) Calculate the value of the rate constant, k , for this reaction. [2]

$k =$



- (iii) A student suggests that the rate determining step for this reaction is:



State, giving a reason, whether this is a possible rate determining step for this reaction. [2]

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- (c) The rate of this reaction can be followed using sampling and quenching.

- (i) Explain what is meant by *sampling and quenching*. [1]

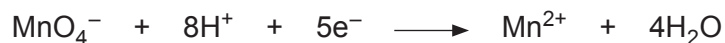
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- (ii) A 5.00 cm^3 sample of the solution was analysed by titration against acidified potassium manganate(VII). The sample required 27.20 cm^3 of manganate(VII) solution of concentration $0.00205\text{ mol dm}^{-3}$ for complete reaction.

Calculate the concentration of Fe^{2+} in the solution. [2]



Concentration of $\text{Fe}^{2+} = \dots\dots\dots \text{mol dm}^{-3}$

